



Advances in atmospheric, oceanic, and coupled models for meteorological forecasting

Muhammad Waqas^{a,b,*}, Usa Wannasingha Humphries^b, Angkool Wangwongchai^b

^a The Joint Graduate School of Energy and Environment (JGSEE), King Mongkut's University of Technology Thonburi (KMUTT), Bangkok, 10140, Thailand

^b Department of Mathematics, Faculty of Science, King Mongkut's University of Technology Thonburi (KMUTT), Bangkok, 10140, Thailand

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ABSTRACT

Numerical Weather Prediction (NWP) has evolved from atmospheric-only formulations to comprehensive Earth System Models (ESMs), driven by advances in model resolution, physical realism, and computational power. While many reviews of ESMs exist, this study discussed the developmental pathway and provided a comparative analysis of atmospheric general circulation models (AGCMs), oceanic general circulation models (OGCMs), coupled atmosphere-ocean general circulation models (AOGCMs), and fully integrated ESMs. This review emphasizes the unique value of coupled models in representing feedback and teleconnections within the Earth system by examining improvements in the simulation of critical climate phenomena. These advances are linked to refined coupling strategies, enhanced parameterization schemes, and high-resolution initiatives. Recent developments in artificial intelligence (AI) and machine learning (ML) are highlighted for their potential to improve predictions, reduce systematic errors, and enable hybrid physics-data models. Ocean models such as Nucleus of European Modeling of the Ocean (NEMO) and Hybrid Coordinate Ocean Model (HYCOM) are assessed for their contributions to mesoscale dynamics, air-sea interactions, and biogeochemical cycling. Nonetheless, persistent challenges remain, including high computational costs, improved cloud and precipitation physics representation, and consistent cross-component initialization. This review concludes by identifying critical research gaps and discussing how AI-augmented physics-based models can support the development of scalable, transparent, and policy-relevant prediction systems.

1. Introduction

With the emergence of numerical weather prediction (NWP) in the mid-20th century, meteorological forecasting has experienced substantial advancements (Lynch, 2008). Richardson introduced the conceptual formulations of NWP in 1922, but the practical application of his techniques was limited by the paucity of computational resources and the primitive state of knowledge of atmospheric dynamics in 1922 (Lynch, 2006).

With the advent of digital computing in the 1950s, it was possible to numerically solve the primitive equations of atmospheric motion, although at a crude spatial and temporal scale (Staniforth and Wood, 2008). These initial NWP models marked a paradigmatic change from the purely statistical forecasting to the mechanistic approach based on the laws of physics, which formed the basis of the later development of

the atmospheric, oceanic, and fully coupled Earth-system models (ESMs) (Ghil, 2019). The improvement of the meteorological forecasting capacities is associated with the understanding that the appearance of atmospheric and climatic phenomena results from complex and nonlinear interactions between the atmosphere, ocean, and cryosphere, as well as between the atmosphere and land surface and biosphere (Palmer, 2019). Therefore, contemporary forecasting systems require models capable of realistically simulating such coupled processes to generate actionable predictions over a broad timescale, including both short-term weather predictions and seasonal to decadal climate predictions (Legg, 2021).

Modern forecasting models can be divided into four closely interrelated categories: atmospheric climate models, oceanic general circulation models (OGCMs), coupled atmosphere-ocean general circulation models (AOGCMs), and global climate models (GCMs) (Grassl, 2000).

* Corresponding author. The Joint Graduate School of Energy and Environment (JGSEE), King Mongkut's University of Technology Thonburi (KMUTT), Bangkok, 10140, Thailand.

E-mail address: muhammad.waqas@kmutt.ac.th (M. Waqas).

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Each category represents a particular facet of Earth-system dynamics, but systemic coupling is fundamental to a realistic representation of feedback processes and teleconnections that are important to practical predictive ability.

Meteorological prediction systems have historically been based on atmospheric models. The primitive models, which had horizontal resolutions of 100 km and crude parameterizations of convection, cloud microphysics, and surface fluxes, were both space and physics limited (Staniforth and Wood, 2008). The further increase of computing power and the creation of advanced numerical algorithms enabled the advancement of spatial quality and the physical accuracy of the simulations. Modern GCMs, such as those used in the Coupled Model Intercomparison Project Phase 6 (CMIP6), often use a horizontal grid spacing between 25 and 50 km, which allows the explicit resolution of mesoscale processes (Eyring et al., 2016; Nowack et al., 2020). Similar progress has been made in modeling cloud microphysics, convective physics, boundary-layer turbulence, and aerosol-cloud interactions, which are crucial to accurately modeling key weather and climate processes (Flato et al., 2014; Randall et al., 2007).

The creation of ESMs is a landmark improvement on the atmospheric-only models. ESMs combine land and ocean biogeochemistry, atmospheric chemistry, terrestrial and oceanic biosphere, and anthropogenic forcing, therefore, allowing the study of complex feedback including aerosol-cloud-radiation interactions and the role of land-use change in surface energy fluxes (Flato et al., 2014; Perkins et al., 2007). These gains notwithstanding, significant uncertainties remain, especially in the cloud formation and convective system structure, which still dominate the forecasting errors and inter-model spread (Stevens and Bony, 2013).

The development of ocean models has highlighted the importance of the ocean in climate regulation, as it has high heat capacity and affects the global energy balance. OGCMs offer a physical description of large-scale circulation, mesoscale eddies, thermohaline processes, and sea ice processes (Griffies et al., 2009). Models like the Nucleus of European Modeling of the Ocean (NEMO) and Hybrid Coordinate Ocean Model (HYCOM) are essential to explain the interaction between the ocean and atmosphere, including the El Niño-Southern Oscillation (ENSO), the Atlantic Multidecadal Oscillation (AMO), and the Pacific Decadal Oscillation (PDO), which vary the weather patterns on a seasonal to decadal scale (Barnston et al., 2019). Issues have remained in the accurate parameterization of mesoscale eddies, vertical mixing, and air-sea flux, which are necessary to enhance the fidelity of the coupled simulations (Fox-Kemper et al., 2019).

The development of AOGCMs has created a methodological breakthrough in the combination of atmospheric and oceanic processes. AOGCMs can produce more lifelike images of the climatic variability and the related teleconnections by explicitly simulating the two-way exchange of momentum, heat, and freshwater between the air and sea interface (Capotondi et al., 2019; Edwards, 2011). Examples of such integrative philosophy include the Community ESM (CESM) and the NOAA Climate Forecast System version 2 (CFSv2), which include complex atmosphere, oceanic, sea-ice, and land-surface components in coherent simulations (Moltmann et al., 2019). Nevertheless, coupled modeling is associated with methodological issues, such as balancing disparate spatial and temporal resolution among components, managing climate drift caused by flux imbalances, and managing transient shocks in initialization through spin-up (Li et al., 2024; Wang, 2019). High-frequency coupling, iterative flux correction, platform-independent software, e.g., OASIS and ESM Framework (ESMF), are examples of such innovations, which have enhanced the stability and reliability of simulations (Gröger et al., 2021; Xue et al., 2020).

NWP models with convection-permitting grid resolutions (<5 km) have also become possible with advances in computational power to explicitly simulate deep convection, tropical cyclones, and mesoscale convective systems, which were previously parameterized with a great

deal of uncertainty (Palmer, 2019; Prein et al., 2015). The use of ensemble forecasting, with its reliance on perturbed first conditions or physics, is now a common practice to quantify the predictive uncertainty, and decision-making has been advanced, with the North American Multimodel Ensemble (NMME) project as a pioneering example, where it is proposed to combine multiple modeling models to increase the robustness of the forecasts (Cornuelle et al., 2014).

The combination of artificial intelligence (AI) and machine learning (ML) has also increased predictive power, and neural networks can help to improve precipitation nowcasting and detect extreme weather. In contrast, physics-informed ML (PIML) can optimize sub-grid parameterizations to improve physical consistency and model performance (Kashinath et al., 2021; Mehrabi et al., 2023, 2024; Waqas et al., 2023). Despite these developments, several challenges remain, including interpretability issues, out-of-sample generalization, and compliance with underlying physical laws. Consequently, a hybrid system combining AI and traditional NWP is a key research direction (Kashinath et al., 2021).

However, there are still significant limitations to operational meteorological forecasting. The construction of ultra-high-resolution and fully coupled ESMs is associated with the trade-offs of complexity, spatial, and ensemble size of the model (El Serafy et al., 2021; Waqas et al., 2024). There are still no complete failures in representing physical and biogeochemical processes, as persistent biases in important variables, such as cloud cover, precipitation, and ocean heat content, are demonstrated (Stevens and Bony, 2013). The search to have a smooth prediction framework of weather and climate scales requires new methods of model startup, stochastic parameterization, and cross-time scale interaction (Bauer et al., 2015).

Similarly, with dimensional increases in observational data offered by satellites, autonomous platforms, and in situ networks, there are opportunities and challenges in model validation and refinement, and it is required to have advanced data management and processing infrastructures (Gao et al., 2023). The future path of meteorological modeling is expected to be based on the joint development of improved physical insights, computing advances, and methodological development of data science.

Although the historical basis is strong, it is now time to synthesize, given the growing need for credible hazard forecasting in the face of increasing extremes. This review thus frames the model developments against the risk management context, focusing on how the innovations in coupled, resolved, and AI integration help target more actionable early warning systems. This review presents a comprehensive analysis of recent developments in meteorological modeling. The primary objectives of this study are to categorize and explain the main types of models, including climate models, oceanic models, and coupled atmosphere-ocean-land models, to identify new technological and algorithmic developments, such as AI-augmented modeling, and to determine the remaining research gaps.

2. Methodology

This systematic review employs an evidence-based methodology to synthesize recent atmospheric, oceanic, and climate modeling advancements, including OGCMs, ESMs, and coupled atmosphere-ocean systems.

A comprehensive literature search was conducted across major scientific databases, including Scopus, Web of Science, ScienceDirect, IEEE Xplore, AGU Publications, and Copernicus GMD, to identify peer-reviewed journal articles, model documentation, and international modeling initiatives. Keywords such as "atmospheric modeling," "ocean modeling," "global climate models," "numerical weather prediction," "coupled climate models," "Earth system modeling," "AI in meteorology," and "climate forecasting systems" guided the search strategy. Only peer-reviewed studies and model evaluation reports detailing model development, application, and intercomparison were retained; non-peer-

reviewed, informal, or tangentially related sources were excluded.

To achieve the objectives of this study, the following research question is designed: (1) What is the historical evolution of these models? What are the core characteristics and differences among atmospheric, oceanic, coupled, and climate models? (2) How are the main climate models classified in operational forecasting across various spatial and temporal scales? (3) What are recent advances? How are new computational and AI techniques influencing the development and performance of coupled models? (4) What limitations and uncertainties are inherent in current modeling approaches?

At the first screening stage, about 155 scholarly articles and modeling reports were identified for thorough analysis. They were systematically categorized into five main categories: (i) AGCMs, (ii) OGCMs, (iii) coupled models, including the AOGCMs and cryosphere biosphere systems, (iv) ESMs, and (v) ML-augmented models, including neural GCMs and PIML. As opposed to a standard narrative or bibliographic review, this paper follows the systematic review protocols, i.e., it follows a systematic and transparent literature identification, selection, categorization, and synthesis procedure. The methodological rigor is evident in several key aspects: the research questions are explicitly defined, internationally recognized scientific databases are utilized, and predetermined inclusion and exclusion criteria are employed to narrow the scope to peer-reviewed, model-based studies.

Moreover, dividing the final corpus of 155 articles into specific model categories (AGCMs, OGCMs, AOGCMs, ESMs, and AI/ML-augmented models) makes studies comparable and reduces the selection bias.

The assessment was based on three important dimensions: physical fidelity, coupling methodology, and performance prediction. Moreover, data assimilation and ensemble post-processing AI-driven tools were addressed, and the problem of generalization, physical consistency, and calibration in extreme conditions was addressed.

To guarantee methodological rigor and reproducibility, grey literature, unpublished preprints, and non-archival model reports were excluded. Formally published evaluation reports of known international modeling centers and intercomparison projects (e.g., CMIP, WCRP initiatives) were kept, as they are more standardized in documentation and peer reviewed. The present synthesis combines historical trends, structural changes, and technical knowledge, thus defining evolution, present opportunities, and prospects of meteorological forecasting models.

3. Historical evolution of modeling

The early 1950s saw the development of numerical meteorology when von Neumann and others at the Institute of Advanced Study made the first computer-based weather forecasts (Lynch, 2006). These early studies formed the foundations of the later construction of three-dimensional AGCMs and OGCMs, which were initially built separately.

In the 1960s, the development of Whole Atmosphere GCMs, which modeled the three-dimensional circulation of the atmosphere at constant sea surface temperatures, allowed the systematic study of climatic responses to external forcings like rising greenhouse gas concentrations (Ghil and Robertson, 2000). Later in the 1960s and 1970s, Manabe and Bryan (1969) further developed the field by fully coupled AOGCMs, which combined AGCMs with dynamically coupled ocean and sea-ice models. This advancement enabled two-way interaction between the atmosphere and the ocean and enabled the emergent representation of such phenomena as the ENSO that could only be represented in fully coupled systems (Manabe and Bryan, 1969). Fig. 1, derived from Tan et al. (2023), presents a complete overview of the historical modeling evaluation (Tan et al., 2023).

Climate modeling. In the 1980s and 1990s, climate modeling developed beyond constrained AGCMs to fully coupled GCMs no longer

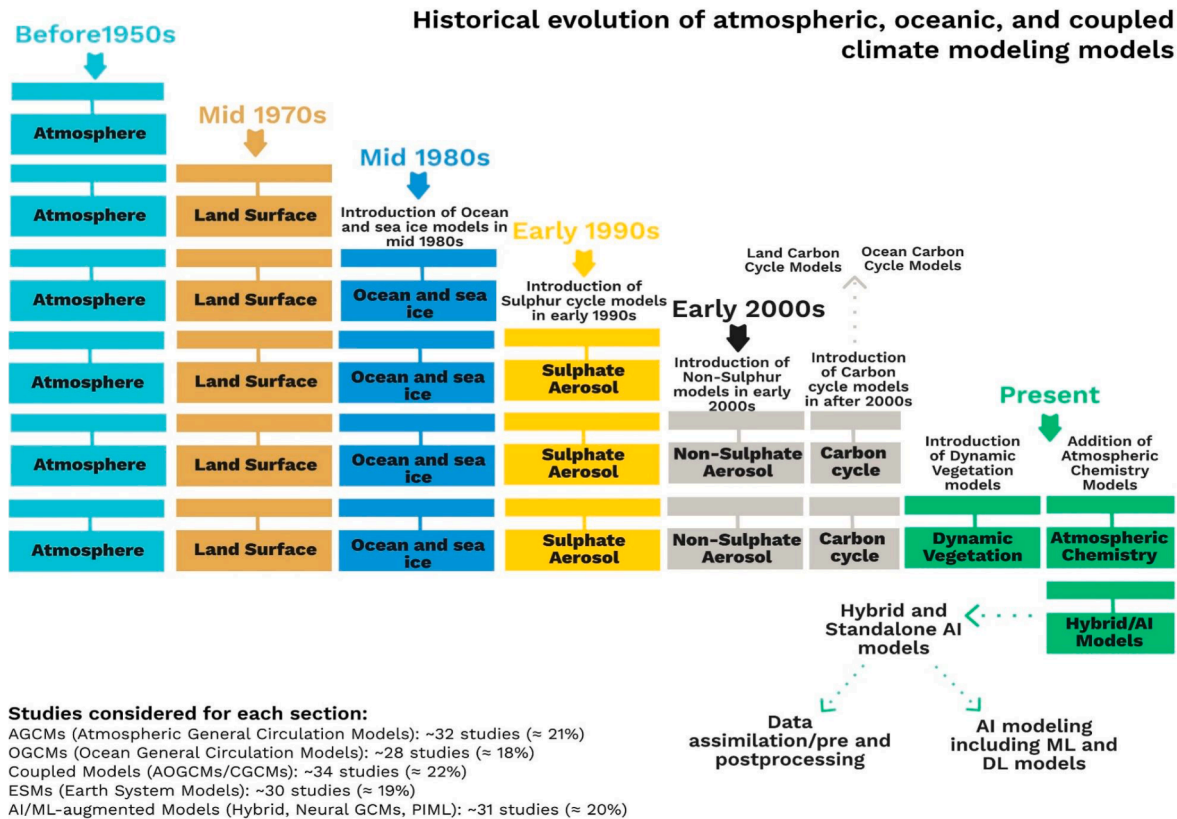


Fig. 1. Historical evolution of atmospheric, oceanic, and climate modeling of the 1950s to the present, with important conceptual and computational developments that led to the development of modern ESMs.

based on artificial flux adjustments, including those used in HadCM3 at the Met Office Hadley Centre. These models exhibited more steady and physically consistent behavior in the atmosphere and ocean than previous models, requiring surface flux adjustments to achieve numerical stability. Meanwhile, community-based projects, such as the Community Climate System Model (CCSM) developed at NCAR and its successor, the CESM, began to include terrestrial biosphere, sea-ice, and more complex biogeochemical processes with each new release (Collins et al., 2006).

At the beginning of the 21st century, ESMs appeared, which combine atmospheric, oceanic, land-surface, sea-ice, and biogeochemical processes, thus allowing a more detailed study of carbon-climate feedback and global biogeochemical cycles (Flato et al., 2014; Perkins et al., 2007). Systematic evaluation of model performance through standardized model intercomparison frameworks (CMIP and its modern form, CMIP6) was used to enhance knowledge of model biases, internal variability, and projections (Eyring et al., 2016; Nowack et al., 2020).

Three synergistic trends led to climate and weather modeling improvements between 2010 and 2020, including increased fidelity of atmospheric and oceanic numerical models, improved data assimilation methods, and increased computational capabilities through exascale and heterogeneous computing systems, through GPU-based systems. These innovations represent a significant leap in the generational dimension, as CMIP6 provides rigorous evaluation schemes for ESMs under various socioeconomic conditions (Eyring et al., 2016). Sub-kilometer scale simulations have become possible in high resolution, but the correct parameterization of physical processes, especially cloud and precipitation dynamics, is a significant challenge (Khairy et al., 2020; Shalf et al., 2010). More recent attempts are progressively based on AI/ML, such as convolutional neural networks and transformer-based models, to model physical processes, reduce biases, interpolate downscaled variables, and optimize sub-grid parameterizations, thus allowing hybrid dynamical-AI modeling paradigms (Dueben and Bauer, 2018; Rasp et al., 2020; Reichstein et al., 2019).

The three principal axes characterizing the development of meteorological modeling are greater physical realism and complexity, smaller spatial resolution, and larger ensemble size. These dimensions, in turn, have significant computational requirements, which require trade-offs depending on the goals of operational centers (El Serafy et al., 2021;

Waqas et al., 2024). The continued bias in models, especially regarding cloud microphysics, precipitation processes, and ocean heat uptake, highlights the necessity of continued methodological innovation (Stevens and Bony, 2013). The models needed to be accurately initialized, dynamically coupled, and stochastically parameterized to achieve the seamless prognostic capability across a wide range of timescales (Bauer et al., 2015), and the growing availability of satellite and in situ observations offers the chance to validate and refine them with the help of the advanced data assimilation (Gao et al., 2023). Furthermore, Table 1 compares ocean, climate, and fully coupled ESMs, detailing their component structures and representative real-world applications.

Furthermore, Fig. 2 illustrates interactions among atmosphere, cryosphere, hydrosphere, biosphere, and human influences, highlighting feedbacks such as circulation, radiation, hydrological cycle changes, and land-atmosphere-ocean coupling within ESMs (Mattoo et al., 2025; Stockwell et al., 2011).

3.1. Atmospheric models

The basis of modern NWP and climate modeling systems comprises AGCMs. These models find a numerical solution to the primitive conservation equations of momentum, mass continuity, thermodynamics, and state equation, usually under hydrostatics and Boussinesq's approximations (Washington and Parkinson, 2005). Vertical resolution is typically applied in either pressure or hybrid sigma-based coordinates, but horizontal resolution has been significantly improved throughout the history of climate studies. The spacing of grids in early climate models was about 200 km, and modern high-resolution operational models can cut the scales below 10 km (Satoh et al., 2014).

Early AGCM models were established in uncoupled form with sea-surface temperature prescribed as a boundary condition, as well as static properties of the land-surface (Collins et al., 2004). Although these configurations allowed sensitivity experiments to be computationally manageable, they had the inherent drawback of restricting the ability to represent coupled feedback between the atmosphere and other parts of the Earth system. Despite these shortcomings, AGCMs are still indispensable to standalone tasks, such as reanalysis datasets (e.g., ERA5, MERRA-2), paleoclimate reconstruction, and specific process studies (Collow et al., 2022).

Table 1

Comparison of ocean, climate, and coupled models with components and real-world applications.

Model Type	Definition	Key Components	Application	Examples
Atmospheric Models	Numerical simulation of atmospheric dynamics and thermodynamics using governing physical equations.	Wind fields, temperature, pressure, humidity, radiation, convection, boundary layer processes, cloud microphysics.	Forecast weather patterns, quantify circulation variability, and support reanalysis, all of which are evaluated using Root Mean Square Error (RMSE), anomaly correlation, and benchmarked against CMIP6 or reanalysis datasets (Liang-Liang et al., 2022; Risser and Wehner, 2020).	ECMWF IFS, NCAR CAM5, MPAS; prediction of tropical cyclones, monsoon onset; ERA5 reanalysis.
Ocean Models	Numerical representation of ocean dynamics and thermodynamics.	Currents, temperature, salinity, sea level, vertical mixing, convection, surface fluxes.	Investigate ocean circulation, heat and carbon transport, and sea-level rise. Model skill is assessed via Root Mean Square Deviation (RMSD) and correlation with ARGO or satellite data, with benchmarks often drawn from CMIP6 HighResMIP (de Souza et al., 2024; Liang-Liang et al., 2022).	AMOC simulations, upwelling studies, and ocean heat transport estimation.
ESMs	Fully coupled models integrating atmosphere, ocean, land, cryosphere, and biosphere components.	Atmosphere, hydrosphere, cryosphere, land surface, biosphere.	Project long-term climate trends and responses to scenarios (RCPs, SSPs); evaluated using multi-metric CMIP6 benchmarks, pattern correlation, and variance explained (Gier et al., 2024).	IPCC projections, global temperature trends, and Arctic ice melting simulations.
Coupled Models	Integration of multiple sub-models to simulate interactions among Earth system components.	Atmosphere + Ocean + Land + Sea ice + biosphere (as applicable).	Examine feedback, teleconnections, and emergent climate phenomena; performance assessed using skill scores, ENSO indices, and CMIP6 evaluation metrics (Beniche, 2024; Dalelane et al., 2023).	CMIP6 experiments, ENSO, monsoon, and decadal climate variability simulations.

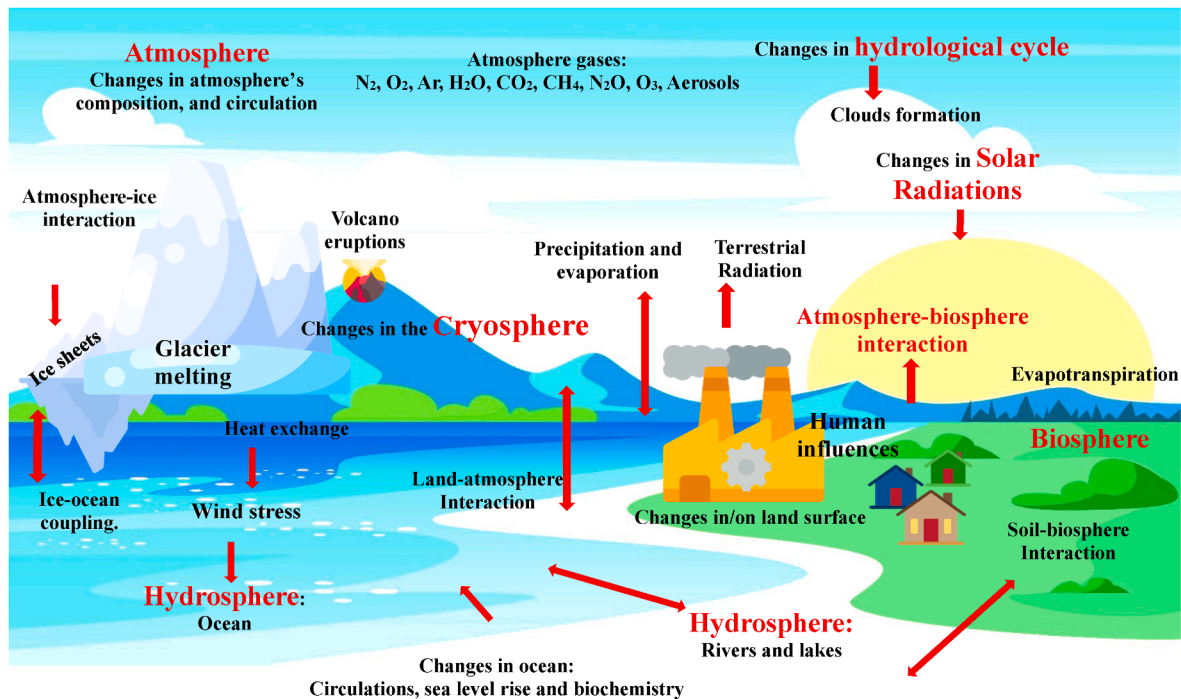


Fig. 2. Overview of important atmospheric processes and interactions with the cryosphere, hydrosphere, biosphere, and human processes within ESMs, showing coupled interactions between energy, water, and biogeochemical cycles (Mattoo et al., 2025; Stockwell et al., 2011).

The physical parameterizations of AGCMs have also improved significantly since 2010. New features of radiative transfer modules include spectrally resolved shortwave and longwave treatments, cloud-aerosol-radiation interactions, and absorption by trace gases (Mlawer et al., 1997). The convection schemes have also been updated with stochastic triggering, convective memory, entrainment and detrainment corrections, and diagnostics of convective organization, which is empirically adapted (Arakawa and Jung, 2011). Better use is made of observational programs and large-eddy simulation outcomes, which limit turbulent fluxes and planetary boundary layer processes (Holtslag et al., 2007).

Advanced modeling systems, such as the Model of Prediction Across Scales (MPAS) developed at NCAR, adopt unstructured mesh refinement to achieve variable spatial resolution of one domain (Skamarock et al., 2008, 2019). It offers a robust level of flexibility because it can refine the areas of interest (e.g., cyclogenesis regions or mountainous topography), without a linear rise in global computational cost. The distorted mesh structure of MPAS, however, can make it difficult to realize numerical implementation and physical parameterizations, and communication overhead can grow when it is coupled with other components of the Earth system (Jablonowski and Williamson, 2011; Ringler et al., 2010). On the other hand, quasi-uniform icosahedral grid, with the ICOSahedral Nonhydrostatic (ICON) model, guarantee less numerical anisotropy and global consistency (Zängl et al., 2015). The ICON grid is scalable to massively parallel systems, and it is easy to interface to ocean and terrestrial components, although its uniform resolution necessitates a large amount of global computational capacity, even in areas where the finer scale is not necessary (Crueger et al., 2018; Zängl et al., 2015). Therefore, MPAS is efficient and flexible at the expense of grid complexity, and ICON is more concerned with numerical regularity and consistency of coupling at the expense of computational cost. The advances in the two systems have led to the enhancement of the representation of intraseasonal variability, the formation of tropical cyclones, and the storm pathways in the midlatitudes.

Recent AI/ML breakthroughs have also added features of sub-grid process emulation, bias removal, post-processing, and faster global forecasting (Laloyaux et al., 2022; Rasp et al., 2020; Reichstein et al.,

2019; Waqas and Humphries, 2025). FourcastNet and GraphCast are architectures that generate skilled global predictions at lower computational expenses (Lam et al., 2023; Pathak et al., 2022).

However, in contrast to physically based models, most of the AI-based methods have issues with reproducibility because they are sensitive to training data, pre-processing decisions, and undocumented hyperparameters (Hutson, 2018; Waqas et al., 2023). Neural architectures can also be viewed as limited in terms of interpretability since they can be black-box, and it is hard to attribute forecast skill to physical processes (McGovern et al., 2019; Rudin, 2019). Out-of-sample generalization could continue to be an issue, especially in a changing climate when historical training data might no longer be the most extreme values in the future (Beucler et al., 2021). These shortcomings demonstrate the danger of over-optimism as well as the need to support hybrid methods that incorporate physical constraints in ML models, which guarantee both robustness and scientific validity (Abramowitz et al., 2019; Rudin, 2019; Sham et al., 2025).

3.2. Oceanic models

Oceanic models are designed to track the spatiotemporal evolution of temperature, salinity, currents, and sea-level height using the primitive equations in the Boussinesq and hydrostatic approximations (Fig. 3) (Zardi, 2024). Grid architecture encompasses structured z-level coordinate systems, unstructured meshes, and those with dynamically adaptive resolution schemes (Kantha and Clayson, 2000). Such operational configurations include the Met Office FOAM (Forecasting Ocean Assimilation Model), which utilizes up to 75 vertical levels of NEMO-coupled components, such as sea-ice models like CICE and complete biogeochemical tracer suites, to supplement these simulations (Feng et al., 2019).

OGCMs solve features of the ocean's mixed layer, thermocline, and mesoscale eddy scales (Zhang et al., 2016). These models cannot be avoided, as they are essential for describing phenomena such as ENSO, the Atlantic Meridional Overturning Circulation (AMOC), and regional ocean variations (Farneti, 2017). That is because the ocean traps over 90 % of the excess heat on Earth and transports it across the basin scale.

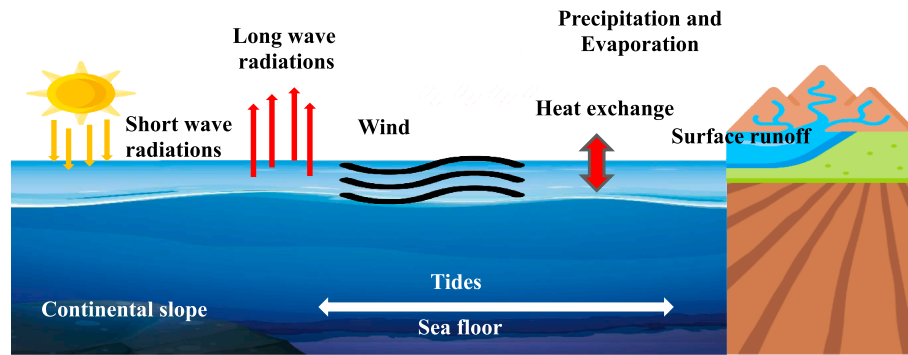


Fig. 3. Ocean model forcings (mainly the sun and earth radiation, precipitation and evaporation, surface run-off, tides, sea bottom interactions, etc.) which together serve to control ocean circulation, ocean thermodynamics, and ocean biogeochemical cycles (Zardi, 2024).

Biogeochemical components, which consider nutrient cycling, biological productivity, carbon export, and ocean acidification, are also critical in explaining ocean heat absorption and its feedback to climatic and weather mechanisms (Honjo et al., 2008). An example is the CESM, which reveals that variations in surface productivity affect ocean heat capacity and air-sea relations through biogeochemically enabled simulations (Li and Striver, 2019).

In addition to climatic processes, ocean models have been crucial in the forecasting of hazards through a specific simulation of storm surge processes and compound flooding. The storm surge prediction and coastal inundation risk assessment are extensively carried out with a coupled ocean-atmosphere model like the Advanced Circulation Model (ADCIRC) that solves the sea-level rise caused by storms and how they interact with coastal bathymetry and topography (Dietrich et al., 2011, 2012). For example, ADCIRC has effectively recreated the extreme surge patterns of Hurricane Katrina in 2005 and Hurricane Sandy in 2012, which gives vital information on the extent of floods and the vulnerability of infrastructure (Dietrich et al., 2012; Fleming et al., 2008). Later, more integrated modeling frameworks, in which oceanic, hydrologic, and hydraulic processes are coupled, have been used to determine the effects of compound flooding, in which heavy precipitation is accompanied by storm surges, and enhances the effects of coastal floods (Moftakhari et al., 2017; Wahl et al., 2015). One of them is the estimation of the risk of compound flooding in the Gulf of Mexico and Southeast Asia, where coupled ocean-atmosphere-hydrology systems enhanced the ability to forecast and manage coastal actions (Bilskie and Hagen, 2018; Wichitnithed et al., 2025). These types of hazard-driven modeling frameworks enhance the usefulness of ocean models in disaster risk mitigation and resilience development in prone coastal areas.

In the field of physical oceanography, two key data assimilation systems—BlackmanV and NEMOVAR—stand out due to their ability to combine satellite and in situ measurements, thereby improving upper-ocean initial conditions and, consequently, enhancing forecast skill in both seasonal prediction and coupled reanalysis applications (Saulter et al., 2020; Whitaker et al., 2008). The necessity to produce proper initial conditions of the heat content and salinity lies in the inherently slow thermal and salinity dynamics at the longer time scales.

3.3. Coupled models

To simulate coupled physical, chemical, and biological systems on Earth, modern climate models integrate atmospheric, oceanic, terrestrial, sea-ice, and biogeochemical components into a comprehensive model (Fig. 4). Centralized couplers such as OASIS-MCT, ESMF, and MCT facilitate the exchange of energy, momentum, moisture, and matter while maintaining computational efficiency and scalability in parallel processing environments (Craig et al., 2012). This architecture supports a two-way flow of information between modules, allowing

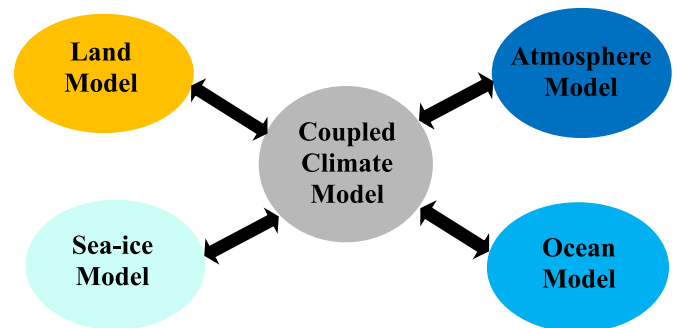


Fig. 4. Schematic representation of a coupled climate model shows the structure of the interactions and feedbacks in the Earth system as the atmosphere, ocean, terrestrial, and sea-ice systems are bound together to interact and form a whole system (Xia et al., 2023).

emergent behavior to arise internally rather than being externally imposed (e.g., via external forcing).

ENSO offers a popular example of coupled model utility. Interactions and feedback between SST, atmospheric convection, and wind were first incorporated into numerical models in the late 1970s and early 1980s, most prominently in the Zebiak-Cane intermediate model. Later efforts, including the TOGA program, refined model performance by incorporating dense observational datasets and optimizing initialization processes (Weickmann, 1991).

In ESMs, the paradigm of coupled systems has long been established as the dominant framework in climate research (Danabasoglu et al., 2020; Hurrell et al., 2013). Heat transport pathways, mesoscale eddies, and feedback processes at the coast can significantly impact atmospheric dynamics, thereby improving model fidelity in phenomena such as tropical cyclones, monsoon onset, and mid-latitude storm tracks. Although two-way interactions can mitigate the climate drift inherent in uncoupled or slab-ocean configurations, coupled models incur high computational costs and require complex calibration procedures. Initialization remains a key issue; coupled data assimilation systems are needed to simultaneously estimate atmosphere and ocean analysis fields, which introduces additional requirements for stochastic innovations and hybrid covariance techniques (Danabasoglu et al., 2020; Fiori et al., 2009; Gröger et al., 2021; Laloyaux et al., 2022; Prestele et al., 2017).

Initialization, especially, has been a perennial problem, as it involves the creation of balanced initial states that represent fields of atmosphere, ocean, land, and cryosphere without causing any spurious shocks or drift. It is handled using coupled data assimilation (CDA), whereby data on several components of the Earth system are ingested in a physically coherent manner, typically by hybrid ensemble-variational methods or weakly coupled data assimilation models (Fiori et al., 2009; Laloyaux

et al., 2022; Penny and Hamill, 2017).

Another major progress has been the recent growth of coupled reanalyses, which combine observational constraints between sub-systems to enhance initialization and calibration. An example of this is the ECMWF CERA-20C reanalysis, which uses a coupled ensemble data assimilation system that builds and ingests twentieth-century atmospheric and oceanic observations in the same time (Laloyaux et al., 2022). Compared to uncoupled reanalyses, which can cause discrepancies across the air sea interface, CERA-20C balances ocean heat content with sea-surface temperature and atmospheric circulation, and thus reduces the biases in initializing the analysis, as well as enhancing the representation of low-frequency variability, including the Atlantic Multidecadal Oscillation and ENSO modulation. The ensemble-based design also gives uncertainty estimates in the form of probabilities, enhancing the robustness of further simulations and projections. The significance of reanalysis products can be illustrated by both hindcasting and attribution research, and the provision of better boundary conditions to initialize climate predictions (Laloyaux et al., 2022; Poli et al., 2016).

A summary of prominent studies that have utilized coupled EDMs to examine fundamental climate processes and regional effects is presented in Table 2. Each case highlights the model-specific potential to advance knowledge and improve understanding of ocean-atmosphere interactions, cryosphere changes, monsoon variability, and projections of climate extremes.

3.4. Comparative insights and core differences

Table 3 summarizes important climate modeling paradigms, including AGCMs and OGCMs that individually model atmospheric and oceanic processes, coupled models (CGCMs/AOGCMs), and ESMs that combine biogeochemical cycles to be more realistic but more complex. Most recently, AI/ML-hybrid models have become available, which provide quicker and effective predictions of short-term extremes, but with less temporal resolution.

4. Classification and description of major models

Operational meteorological forecasting is based on climate, ocean, and coupled model ecosystem models, customized for different spatial and temporal regimes. Among them, CMIP6, WRF, NEMO, HYCOM, CESM, and CFSv2 are characterized by the sophistication of their methods and various coupling approaches. This section examines their structural design, resolution capabilities, and coupling frameworks.

4.1. CMIP6 (HighResMIP experiments)

CMIP6 is the current standard protocol of global climate modeling. In CMIP6, the High-Resolution Model Intercomparison Project (HighResMIP) is a systematic investigation of the effect of increased horizontal resolution on climate models, with both standard (~ 100 km atm, $\sim 1^\circ$ ocean) and high-resolution (~ 25 km atm, $\sim 0.25^\circ$ ocean) configurations (Chen et al., 2022; Kodama et al., 2020; Roberts et al., 2019).

HighResMIP specifies a layered experimental scheme:

Tier 1: forced atmosphere-only simulations (1950–2014) with prescribed SST and sea-ice fields at 0.25 resolution.

Tier 2: coupled atmosphere-ocean runs (1950–2050), spin-ups, and historical and future projections.

Tier 3: future atmosphere-forced runs (2050 and beyond to 2100).

Such setups allow for the assessment of the effects of resolution on biases and variability. For example, HadGEM3 GC3.1 was configured with a range of 130 km- 1° to 25 km- $1^\circ/12^\circ$ resolution, showing decreased radiation field bias and enhanced ocean dynamics at higher resolutions (Roberts et al., 2019).

HighResMIP promotes better representation of small-scale to atmosphere-scale processes, including convection systems, regional

Table 2

Applications of major climate models across regions and processes, summarizing model name, scope, key findings, and references.

Model	Application area	Key findings	Reference
EC-Earth3	ENSO Prediction	Improved representation of tropical Pacific feedback, notably reducing the persistent "double ITCZ" bias.	Wyser et al. (2021)
GFDL-CM4	Atlantic Meridional Overturning Circulation (AMOC)	Projected a 34 % weakening of AMOC by 2100, attributed to increased freshwater input from melting ice sheets.	Zhang et al. (2023)
MPI-ESM1.2	Indian Monsoon Variability	Identified links between monsoon variability, aerosol forcing, and Indian Ocean Dipole interactions.	Pall et al. (2024)
NorCPM1 (Norwegian Climate Prediction Model)	Decadal Climate Forecasting	Demonstrated predictive skill for North Atlantic SSTs 5–10 years in advance using coupled data assimilation.	Nair et al. (2024)
UKESM1 (UK Earth System Model)	Amazon Rainforest Tipping Points	It showed that continued deforestation could reduce regional rainfall by up to 40 %, potentially triggering irreversible ecosystem drying.	Mulcahy et al. (2022)
CMCC-CM2	Mediterranean Climate Extremes	Project a $+3^\circ\text{C}$ increase in SSTs by 2100, intensifying heatwaves and marine heatwaves in the region.	Nicoli et al. (2023)
FIO-ESM v2.0	PDO	Coupled ocean-atmosphere processes were identified as the primary driver of PDO phase transitions.	Shu et al. (2024)
AWI-ESM (Alfred Wegener Institute ESM)	Antarctic Ice Sheet Dynamics	Research has revealed that Southern Ocean warming is accelerating basal melting, heightening the risk of ice shelf collapse.	Ackermann et al. (2023)
CNRM-ESM2	Sahel Drought Mechanisms	Linked long-term drought episodes to Atlantic Multidecadal Variability and regional land-atmosphere feedback.	Li and Wang (2022)

precipitation extremes, and tropical cyclones, and increases the quality of seasonal climate and extreme-weather forecasts. HighResMIP temporal outputs are available at sub-daily to monthly frequencies, reflecting synoptic-scale variability and longer-term trends (Chen et al., 2022; Kodama et al., 2020).

The coupling of HighResMIP is necessarily constricted, and the components of the Earth system are fully integrated, exchanging fluxes at high frequencies (on an hourly basis). Importantly, it has adopted minimal returning at scaling levels, leaving resolution as the significant variable (Moreno-Chamarro et al., 2025).

4.2. CESM

CESM, developed by NCAR, is a fully coupled, comprehensive Earth system model that combines atmosphere (CAM), ocean (POP), land (CLM), and sea-ice (CICE) components (Danabasoglu et al., 2020). The high-resolution CESM1.3 model runs at a horizontal resolution of 0.25 in its atmospheric CAM5 and land CLM4 components. The ocean (POP2) and sea-ice (CICE4) components are set at 0.1, and the model employs

Table 3

Comparative characteristics of climate modeling frameworks, detailing roles, resolution, complexity, coupling, strengths, and limitations.

Model	Core role	Resolution	Complexity	Coupling	Strengths	Limitations	References
AGCMs	Simulated atmospheric processes	~50–200 km, sub-daily to seasonal	Medium	None or slab ocean	High-resolution atmospheric dynamics, radiative feedback	No ocean coupling, drifting on decadal scales	(Arakawa and Jung, 2011; Collins et al., 2004; Collow et al., 2022; Mlawer et al., 1997; Satoh et al., 2014; Washington and Parkinson, 2005)
OGCMs	Ocean circulation, thermodynamics	~10–100 km, daily to decadal	High	None or forced	Detailed thermal & baroclinic processes	No feedback to the atmosphere	(Farneti, 2017; Honjo et al., 2008; Kantha and Clayson, 2000; Li and Striver, 2019; Zhang et al., 2016)
CGCMs/AOGCMs	Atmosphere–ocean climate interaction	~50–200 km, daily to centennial	High	Fully bidirectional	ENSO, climate variability, transient climate	Computationally expensive, initialization drift	(Danabasoglu et al., 2020; Hurrell et al., 2013; Kumar et al., 2022; Weickmann, 1991)
ESM	Integrated climate biogeochemistry	~50–200 km, seasonal to centennial	Extremely high	Full multi-sphere	Carbon cycle, feedback mechanisms	Complex calibration, heavy computation	(Danabasoglu et al., 2020; Hurrell et al., 2013; Lalouaux et al., 2022)
AI Hybrid/ML Models	Accelerated forecasting	Coarse (~10–100 km), days to sub-seasonal	Variable	Hybrid	Fast ensembles, improved extreme event skill	Resolution bottlenecks, data dependence	(Lalouaux et al., 2022; Rasp et al., 2020; Reichstein et al., 2019; Waqas and Humphries, 2025)

hybrid vertical coordinates (Danabasoglu et al., 2020).

At this high resolution, CESM improves the representation of meso-scale ocean features (e.g., eddy interactions) and atmospheric processes (e.g., cloud regimes and boundary layer processes). Time outputs are between the hourly and monthly data, allowing for analyzing weather extremes, diurnal cycles, and seasonal to decadal interactions (Ma et al., 2016; Small et al., 2019).

The components of CESM are strongly intercoupled via a coupling layer (e.g., CPL7), and momentum, heat, and freshwater fluxes are exchanged hourly to inter-hourly to ensure dynamical consistency and physical consistency between modules (Danabasoglu et al., 2020).

4.3. NEMO

NEMO is a modular, flexible ocean model used extensively worldwide (Madec, 2015). At regional scales, it serves as the Nucleus of European ocean modeling. Its grid resolutions range from ~1° in global, lower-resolution systems to 0.25° or better (~0.1°) in regional, coastal, or global eddy-resolving systems (Siddorn, 2016).

Vertically, NEMO adheres to z-level coordinates with various levels of depth. Outputs are generated hourly to daily, ideal for capturing coastal dynamics, tidal variability, and mesoscale phenomena (Siddorn, 2016).

The atmospheric model coupling occurs in loose or semi-tight modes. In decoupled regional systems, NEMO can interchange fluxes daily without running time coupling. In frameworks such as WRF-NEMO, a semi-tight coupling is used—sedentary model runtimes exchange instantaneous boundary fluxes but otherwise run quite differently at each time Step (Breivik et al., 2015; Couvelard et al., 2020).

4.4. HYCOM

HYCOM employs a dynamic vertical grid, with z-levels in deep water

and isopycnic/tracked layers in the open ocean. HYCOM is designed to deliver eddy-resolving resolutions (~0.08°–0.25°, or 10 km) that provide detailed insights into mesoscale ocean physics, including fronts, eddies, and regional circulation (Chassignet, 2004). It is embedded in tightly coupled systems, such as CESM-HYCOM, and allows high-frequency coupling, with ocean dynamics updated relative to atmospheric forcing every few hours (Bozec and Chassignet, 2020). This close integration leads to expertise in modeling processes such as air-sea fluxes, mixed-layer processes, and seasonal variability.

4.5. WRF

The WRF is the regional NWP workhorse, featuring two meteorological dynamical cores (ARW and NMM) and an extensive set of physics options. WRF allows grids to be nested at various resolutions, ranging from 50 km (mesoscale) to 1 km (convection-permitting) (Gallus Jr and Bresch, 2006; Storm and Basu, 2010). WRF is most used in regions of 100 km and nested local sub-regions, with temporal frequencies of output ranging from hourly to sub-hourly (e.g., 515 min in convection studies). Larger-scale models or analyses supply lateral boundary conditions (Kittel, 2021). WRF is compatible with high-fidelity coupling schemes. Newer releases (e.g., WRF 4.6.0) incorporate non-intrusive multiscale interfaces to enable seamless coupling with ocean models, such as NEMO or HYCOM, land-surface models, wave models, or chemistry modules. Coupling is usually semi-tight, exchanging fluxes at each timestep (minutes to hours), but executing components run independently (Masson et al., 2024).

4.6. CFSv2

CFSv2 is an operational coupled model developed by NOAA/NCEP for making sub-seasonal to seasonal (S2S) predictions (Riddle et al., 2013). It is a configuration that couples the atmosphere (GSI-initialized

Table 4

Resolution and coupling characteristics of major climate models, specifying atmospheric–oceanic scales, vertical coordinates, and temporal frequency.

Model	Atmosphere resolution	Ocean resolution	Vertical coordinates	Temporal resolution	Coupling
CMIP6	~100 km → 25 km	~1° → 0.25°	Hybrid/z-level	6 h – monthly	Tight
CESM	~100 km → 0.25°	~1° → 0.1°	Hybrid	Hourly – monthly	Tight
NEMO	–	0.25°–1° (regional)	z-level	Hourly – daily	Loose to Semi-Tight
HYCOM	–	0.08°–0.25°	Hybrid	Hourly – daily	Tight
WRF	1–50 km (regional)	–	Eta-like (terrain)	Minutes – hourly	Semi-Tight
CFSv2	0.5°–1°	0.5°–1°	Hybrid	3-h (atm), daily (OCN)	Tight

Grid Point Statistical Interpolation), ocean (GFDL MOM4), land (Noah LSM), and sea ice (CICE) models, all at grids of $\sim 0.5^\circ$ – 1.0° resolution (Srivastava et al., 2017).

CFSv2 fluxes are exchanged hourly or less between modules, and their coupling is tight because of coherent dynamical evolution. Output is stored at 3-hourly (atmosphere) and daily (mean ocean) intervals, enabling stringent S2S analyses (Riddle et al., 2013; Tian et al., 2017).

The strong coupling between the atmosphere and ocean enables the accurate representation of ocean-atmosphere interactions crucial in seasonal forecasts, such as ENSO, monsoon processes, and teleconnections driven by the sea (Camberlin et al., 2001). Furthermore, Table 4 summarizes these models' spatial and temporal comparisons.

4.6.1. Global climate models: CMIP6 & CESM

CMIP6 and CESM are run in global domains with horizontally resolved scales of 100 to 25(km) in the atmosphere and 1 to 0.1(degrees) in ocean layers. Their hybrid vertical coordinates and hourly coupling also make them robust in representing climate processes, which are necessary in simulations to support IPCC-quality climate projections and feedback studies (Lin et al., 2020; Simpson et al., 2020).

4.6.2. Ocean-focused systems: NEMO & HYCOM

Built to model the ocean at high resolutions, NEMO and HYCOM excel in mesoscale and boundary current dynamics, which are essential for modeling sea ice, maritime influences, and coastal weather conditions (Hyun et al., 2022). The eddy-permitting features of HYCOM at a resolution of <0.1 are suitable for the dynamically essential ocean structures. Whereas HYCOM employs tight coupling in global systems, NEMO can be used independently or semi-coupled in regional systems (Bosilovich et al., 2012).

4.6.3. Regional and seasonal weather systems: WRF & CFSv2

WRF offers convection-permitting regional prediction at vast scales, with nesting and advanced physics required to predict extreme weather and conduct localized research. Coupling through new interfaces promotes integrations among the ocean, atmosphere, and land (Mohanty and Mohanty, 2023). CFSv2 fills the time gap between weather and climate prediction, providing operational forecast products at the sub-seasonal to seasonal time scales (Tian et al., 2017).

4.6.4. Coupling strategies

WRF provides highly scalable regional predictions at convection-permitting grid scales, with nesting and improved physics necessary to produce accurate extreme weather forecasts and localized studies. New interfaces support ocean-atmosphere-land integrations with coupling (Gallus Jr and Bresch, 2006; Storm and Basu, 2010).

CFSv2 bridges the temporal gap between weather forecasts (days to weeks) and climate forecasts (months to seasons), delivering operational forecast products at sub-seasonal to seasonal scales (Wang et al., 2010). It is a 0.5 – 1°C resolution grid with tight coupling to enable interactive ocean-atmosphere signals in sub-seasonal to seasonal (S2S) forecasting skill (Peng et al., 2013).

Tight coupling is most advantageous for simulating reciprocal processes, such as SSTs, cloud-radiation interactions, ocean currents, and ice dynamics, which are crucial for reproducing detailed weather and

climate conditions (Frenkel et al., 2015). However, this approach comes at the expense of demanding computational resources. The comparative coupling paradigms draw the differences in the type of execution and frequency of data exchange between loose and tight coupling methods (Table 5). These categories depict the balance of computational efficiency and physical integration of components in operational climate models.

5. Recent trends and innovations

5.1. Advancements in modeling systems

The capabilities of climate and weather forecasting models have been increased to accommodate the recent developments in high-performance computing (HPC). The shift of old-fashioned CPU-based clusters to heterogeneous systems with Graphics Processing Unit, Tensor Processing Unit, and other AI accelerators has enabled ultra-high-resolution simulations at global scales. The developments allow operational centers to reach solutions of about 1 km, which was impossible in the global models before (Ejarque et al., 2022; Jha et al., 2022; Subich, 2025). The operational centers have used these computational improvements to enhance model resolution. An example is the European Centre of Medium-range Weather Forecasts (ECMWF) High-Resolution Ensemble System (HRES), which has a horizontal resolution of about 9 km and 137 vertical levels, which are completely coupled to ocean, sea ice, and wave models using the Integrated Forecasting System (IFS) and 4D-Var data assimilation (Rabier, 2025). On the same note, the National Oceanic and Atmospheric Administration (NOAA) Global Forecast System (GFS) has been modified to a Finite-Volume Cubed-Sphere (FV3) dynamical core, which doubles the vertical resolution to 127 levels and adds the Wave Watch III model, which improves the representation of tropical cyclones and mesoscale precipitation (Zhang et al., 2024).

5.2. Integration of AI/ML in modeling

AI/ML in climate models has significantly improved model calibration, post-processing, and ensemble forecasting (Waqas et al., 2024). Hybrid systems incorporating ML methods in the data assimilation loop have demonstrated potential in correcting model bias, downscaling output, and calibrating ensembles (He et al., 2023). For example, AI/ML models like U-Net have been used to enhance sub-seasonal precipitation forecasts in East Asia, which is better than the conventional regression-based models (Liu et al., 2024).

FourCastNet (Pathak et al., 2022), Pangu Weather (Bi et al., 2022), GraphCast (Lam et al., 2023), FuXi 2.0 (Zhong et al., 2024), FengWu, and AIFS (Lang et al., 2024) provide quick, deterministic forecasts, allowing large-member ensembles (e.g., ~ 100 members in FourCastNet2) to be realized at negligible computational cost. It makes ensemble forecasting democratic for regions with poor HPC resources. Such models employ 3D Transformers and Graph neural networks as an advanced architecture to process intricate meteorological information effectively. It is important to note that FuXi 2.0 is an advanced ML model that offers 1-hourly global weather predictions and a full range of necessary meteorological variables and, therefore, increases its applicability in numerous industries, such as wind and solar energy, aviation,

Table 5
Comparison of coupling paradigms in operational climate modeling based on execution type and data exchange.

Coupling Paradigm	Description	Data Exchange Frequency	Execution Type	Examples
Loose Coupling	Models run independently; exchange boundary data asynchronously	Daily or less frequently	Independent execution	NEMO-based standalone ocean forecasts
Semi-Tight Coupling	Components run concurrently; exchange fluxes each internal time step	Minutes to hours	Concurrent but separate execution	WRF–NEMO regional setups
Tight Coupling	Fully integrated execution with frequent bidirectional flux exchanges	At coupling intervals (persistent)	Integrated execution	CMIP6 HighResMIP, CESM, HYCOM-in-CESM, CFSv2

and marine shipping (Larsson et al., 2025; Zhong et al., 2024).

Moreover, physics-constrained ML models, which apply conservation laws during training, have improved the predictability and strength of forecasts (Kashinath et al., 2021). These models are called Neural GCMs, which combine various differentiable solvers and learned parameterizations that approximate the performance of full GCMs, offering large computational speedups (Kochkov et al., 2024).

5.3. Ensemble forecasting and post-processing

Ensemble forecasting is still an element of probabilistic weather prediction. Conventional models like the NMME, TIGGE archive, and Global Ensemble Forecast System Version 12 (GEFSv12) use several dynamical models to sample initial conditions and model-based uncertainty (Gneiting and Raftery, 2005). Recent developments have supplemented these ensembles with post-processing methods based on ML (Waqas et al., 2023). For example, the bias-correct and downscale ensemble precipitation fields have been trained using adversarial learning models such as Corrector Generative Adversarial Networks (GANs) to enhance the accuracy of the forecasts (Larsson et al., 2025).

Also, AI-based models are directly involved in ensemble generation. FourCastNet, Pangu Weather, GraphCast, FuXi 2.0, FengWu, and AIFS are models that can deliver high-speed deterministic forecasts, so that large-member ensembles can be realized at insignificant computational cost. It democratizes ensemble forecasting, making it available to areas with minimal HPC resources (Larsson et al., 2025; Zhong et al., 2024).

5.4. Limitations and challenges of AI

Although the application of AI/ML integration has improved, several challenges remain (Waqas et al., 2024). Empirical research indicates that AI models are more likely to obscure predictions when the lead time is longer and are worse performers in extreme events, especially those related to extratropical and high-latitude areas (Bonavita, 2024). Such constraints indicate that additional studies should be performed to increase the strength and effectiveness of AI-based predictions.

Scientific roadmaps focus on the necessity to tackle the gaps in interpretability, the problem of generalization in the context of future climate, and the issues of physical consistency of AI models (Larsson et al., 2025). There are attempts to address them by including physical constraints, building probabilistic neural ensemble models, and improving ML systems built on fluid dynamics.

The combination of HPC-based resolution improvement with AI/ML innovations is paving the way for creating hybrid AI NWP systems. These systems are designed to combine the benefits of high-resolution simulations with the efficiency and speed of AI models. High-resolution 4D atmospheric reconstructions (20CRv3 and ERA5) allow the training of AI systems to have a strong climatological basis, allowing the accurate modeling of climatology and enabling bias-sensitive learning (Lang et al., 2024).

The advent of AI-first forecasting models, which are competitive and even superior to standard NWP models, at a fraction of the associated cost, represents a paradigm shift in meteorological forecasting (Dewitte et al., 2021). Furthermore, Table 6 compares leading AI-based weather forecasting models, highlighting their architectures, performance advantages over conventional systems, and domains of applicability. Such systems already influence the forecasts of seasonal monsoons, cyclones, and ENSO. Nevertheless, overcoming the issues associated with extreme events, interpretability, and physical fidelity that will allow advancing AI-based forecasting systems is essential.

6. Limitations, challenges, and future directions

6.1. Model complexity and computational constraints

Higher spatial resolutions or more intricate parameterizations of the

Table 6

Comparative evaluation of AI-based weather forecasting models.

Model	Key features	Performance highlights	Domains of applicability
Pangu Weather	Transformer-based 3D DL architecture	Demonstrates approximately 10 % lower RMSE in Z500 compared with IFS, while extending cyclone track lead times by ~12 h. Produces global forecasts in ~1.4 s, which is nearly 10,000 times faster than conventional models (Bi et al., 2022)	Particularly effective for global-scale deterministic forecasts and tropical cyclone tracking, with the strongest gains in mid-latitude circulation and synoptic-scale prediction.
FourCastNet	Adaptive Fourier Neural Operator enabling rapid surrogate modeling	Matches or surpasses IFS performance for short-term forecasts of global circulation and fine-scale precipitation, delivering a one-week forecast in less than 2 s (Pathak et al., 2022)	Best suited for short-to medium-range global and regional precipitation forecasting, especially over East Asian monsoon regions where fine-scale convective structures dominate (Liu et al., 2024)
GraphCast	Graph-based neural network capturing non-local spatial dependencies	Outperforms Pangu Weather in RMSE and anomaly correlation, showing superior skill for convective processes and tropical systems (Lam et al., 2023)	Provides the most significant improvements for convective and tropical weather prediction, including mesoscale storm dynamics and tropical cyclone evolution, where non-linear interactions are critical.
FuXi 2.0	Coupled atmosphere–ocean ML model with hourly global updates	Surpasses ECMWF HRES in wind power forecasts and tropical cyclone intensity prediction, while providing full meteorological variables at sub-hourly scales (Zhong et al., 2024)	Most advantageous for coupled atmosphere–ocean forecasting, with strengths in tropical cyclone intensity estimation and renewable energy applications such as wind and solar power (Larsson et al., 2025)
FengWu, FuXi, and AIFS (second-generation AI models)	Multimodal encoder–decoder architectures with cascade and graph-based attention mechanisms	Demonstrate improved integration of physical constraints with AI, strengthening predictions across multiple variables and outperforming conventional baselines in selected benchmarks (Lang et al., 2024)	Show the better performance for emerging applications requiring multivariable Earth system prediction, especially extreme event forecasting such as ENSO dynamics, floods, and compound hazards (Dewitte et al., 2021)

models are likely to increase the computational requirements. To demonstrate this, a 10 km $\times$ 1 km grid cell can be reduced to 1 km $\times$ 1 km, multiplying the calculations by approximately 1000 (Buckeridge, 2010). This expansion limits the operation centers to coarse spatial resolution in atmospheric and ocean models. Extreme meteorological events, such as typhoons, have been simulated with high resolution, proving more accurate forecasting (MacLeod et al., 2021). However, scaling such simulations to a global or multidecadal scale is too costly. Some of these issues can be addressed by exascale computing, yet even the most advanced systems take millions of CPU hours to simulate km-scale, coupled ESMs (Beuclet et al., 2023).

As a possible alternative, data-driven systems, particularly ML-based systems, have become viable. In one instance, such as FuXi S2S, it has been shown that ML technologies can make similar or better sub-seasonal predictions than traditional ones up to 42 days ahead at a significantly reduced computational rate (Chen et al., 2024). Global daily forecasting can now be made in just a few seconds compared to hours. Nevertheless, these ML models also have issues, including interpretability issues, calibration issues, and the inability to be used in unfavorable weather conditions (Bonavita, 2024; Van Zyl et al., 2024; Wu and Xue, 2024).

The latter trade-off is manifested in that purely physical models are incredibly fidelities, yet use enormous computational resources, whereas data-driven approaches are fast yet lack interpretability and resistance to outliers. The ML Hybrid and Physical models offer good prospects. It is stated that ML can accelerate the parameterizations or downscaling schemes with a minimum loss of physical consistency (Hess et al., 2025; Mardani et al., 2025).

6.2. Downscaling and regional model integration

Global coupled climate models have a low resolution, which often makes them incapable of capturing small-scale processes, including orographic precipitation, local convection, and urban microclimates. Dynamic and data-driven downscaling methods are to be employed to

Table 7
Strategic approaches for advancing climate modeling.

Strategy	Description
Hybrid Architectures with Physical Constraints	Design ML-informed modules that respect physical laws (e.g., conservation laws, thermodynamics) using techniques like PINNs, diffusion-based correction layers, or ML-assisted parameterizations.
Dynamical Generative Downscaling Frameworks	Utilize diffusion or transformer networks to efficiently downscale global ensemble outputs while preserving physical coherence through reference to dynamical simulations.
Standard Open, Interoperable Hybrid ESM Platforms	Promote initiatives like Destination Earth (DestinE) to create modular, transparent, and interoperable ESM platforms, allowing model benchmarking and greater public accessibility.
Quantify and Communicate Model Limitations and Uncertainty	Use ensemble ML methods, diffusion-based uncertainty quantification, and Decision Making under Deep Uncertainty (DMDU) frameworks to inform policymakers on model limitations and risks.
Exascale-Enabled Regional Coupled RCMs	Support high-throughput computing investments to run high-resolution RCMs like RASM or MPAS for better predictions of extremes and variability in key regions.
Bridge Research and Policy with Digital Twins	Implement digital twin projects like Destination Earth to offer real-time scenario testing and decision support for climate risk and infrastructure planning.
Facilitate Cross-Sectoral Collaboration and Training	Foster interdisciplinary collaboration among meteorologists, oceanographers, data scientists, and policy analysts, with dual training in physical climate models and ML techniques.

fill this resolution gap (Kendon et al., 2021).

Dynamical downscaling uses nested RCMs or limited-area models (LAMs) that use coarse global data that has been assimilated. For example, surface fluxes and atmospheric conditions that a global model cannot resolve may be represented by the Regional Arctic System Model (RASM), and there are no biases in the lower troposphere (Robertson, 2016). Nevertheless, decadal, multicomponent simulations are still prohibitive regarding their computational cost.

Statistical and AI/ML downscaling have significantly progressed (Waqas and Humphries, 2025). The latent and generative diffusion models are now a competitive alternative to dynamical methods. To illustrate, a study showed that latent diffusion decreased the input fields of ERA5 to a 2 km resolution, which was much better than U-Net and GAN baselines and required much less computational effort (Hess et al., 2025; Mardani et al., 2025; Tomasi et al., 2025). On a similar note, a generative diffusion model simulated km-scale phenomena, like typhoons and cold fronts, over Taiwan, but the lack of calibration of uncertainty is an open challenge (Mardani et al., 2025).

Dynamical generative downscaling is a hybrid system comprising dynamical downscaling and ML refinement (Lopez-Gomez et al., 2025). Integrated solutions make large-scale climate ensembles capable of realistic uncertainty quantification, without the large computational cost that full RCM suites impose on each member. Besides, transformer-based and zero-shot consistency models reduce the need to retrain across multiple ESMs (Curran et al., 2024).

Making high-resolution forecasts with less effort is possible by combining regional and ML-based downscale strategies. However, to be healthy, models should be able to deal with uncertainty quantification, physical consistency, and multimodal extremes (Bonavita, 2024; Hess et al., 2025).

6.3. Integrated earth system models

Climate variability and extremes cannot be accurately forecasted without including atmospheric, oceanic, terrestrial, sea-ice, and biosphere-geochemical components. These couplings are studied using ESMs, and such phenomena as the double Intertropical Convergence Zone bias and cryosphere-atmosphere interactions cannot be simulated using uncoupled models (Beuclet et al., 2023).

Such European projects as Destination Earth (DestinE) are set to create a planetary digital twin by applying fully coupled models to utilize real-time in situ measurements and satellite observations to generate policy-relevant applications and disaster response (Nativi and CRAGLIA, 2020). At the same time, global facilities such as ECMWF are concerned with model coupling, data assimilation developments, and reanalyses, such as ERA5, on which credible forecast products are built (de Rosnay et al., 2022).

The coupling value is also increased with physical hybrid methods involving AI. New ESMs increasingly incorporate AI into physical hybrid designs, focusing on maintaining physical constraints, explicit expression of inter-variable dependence, and enhanced extreme event fidelity (Beuclet et al., 2023).

However, challenges persist. High-resolution simulations are still computationally expensive; the components of AI/ML may be unreliable because of inconsistency in physics (Bonavita, 2024). The hybrid formulations must be benchmarked strictly, and best global integration relies on standard, open architectures (Beuclet et al., 2023; Curran et al., 2024).

6.4. Data quality, representativeness, and uncertainty

In addition to the limitations of computations and models, the representativeness and quality of observational data are paramount to the reliability of the forecasts and projections (Beuclet et al., 2021; Mehrabi et al., 2024). The model initiation and validation process require regular, bias-corrected, and quality-assured observational

inputs, but the existing data streams are not evenly distributed spatially and temporally. Although satellite retrievals are advantageous in being global, they can be compromised by calibration drift, retrieval algorithm errors, and the inability to resolve near-surface and cloud-layer processes (Dee et al., 2011). On-ground data can be more accurate in small scales but are unavailable in oceans, polar regions, and parts of the Global South, which can cause systematic representativeness errors in reanalyses and forecast initialization (Lan et al., 2023).

Such uncertainties are also applied to the models' biases, especially of variables like soil moisture, aerosols, and precipitation extremes (Lee et al., 2017). Quality assurance and harmonization structures (as utilized in the reanalysis of ERA5) have improved the assimilation of heterogeneous data streams. However, it is still difficult to ensure that high-frequency and non-traditional data streams generated by citizen science, the Internet of Things, and commercial satellites meet rigorous reproducibility and traceability criteria (Hersbach et al., 2020; Yang et al., 2022). Systematic curation of data, quantification of uncertainty, and measures of representativeness must be performed to make Earth-system simulations reliable and policy-relevant (Waqas et al., 2024). Table 7 outlines strategic approaches for advancing climate modeling, spanning hybrid ML-physics architectures, generative downscaling, and interoperable ESM platforms.

7. Conclusion

The history of meteorological and climate modeling has moved beyond early AGCMs and OGCMs, which did not explicitly model interactions between components of the Earth system, to coupled AOGCMs and fully integrated ESMs, which include important interactions, e.g., ocean-atmosphere heat fluxes, land-biosphere interactions, and cryosphere interactions. Such developments have significantly advanced the simulation of large-scale processes, such as ENSO, AMOC variability, and Arctic amplification, and current systems, such as CESM, CMIP6, WRF, and CFSv2, capture finer spatiotemporal processes across multiple spheres of the Earth. However, these fidelity gains are very costly to compute, highlighting the timeless trade-off between model accuracy and operational feasibility.

Recent AI/ML systems, including FourCastNet, Pangu-Weather, and GraphCast, have shown promise to run simulations faster, with better bias correction, and in better physical consistency to give real-time predictions. Hybrid methods, especially physics-informed ML, promise to incorporate governing constraints in data-driven architectures and hence interpretability and scientific strength. Despite these developments, there are several important challenges. Continued biases in major climate variables do not allow confidence in projections, and global simulations with high resolution are both computationally limited and model-scaling limited. The skill of sub-seasonal to seasonal predictions remains below that of daily and decadal scales, limiting readiness to new climate extremes.

Moreover, the description of the hazards of compounds, e.g., heat-waves parallel with droughts or cascading hydrometeorological risks, is not yet well-developed. The systematic incorporation of ML into data assimilation schemes is also an unsolved problem, especially in the question of striking a balance between physical realism and data-driven flexibility. Also, cloud microphysics and parameterizations of precipitation are a significant area of uncertainty, and standardized multimodel intercomparison protocols are yet to be standardized.

Future development should prioritize hybrid modeling systems, generative downscaling methods, and standardized ESM frameworks explicitly linked to operational forecasting and policy-driven applications. Strengthening these connections will enable integrated and scalable prediction systems that directly support hazard monitoring, disaster preparedness, and evidence-based decision-making. These knowledge gaps need to be addressed to promote climate resilience, enhance disaster risk reduction measures, and sustainable planning in the face of increasing and compounding climate-related hazards.

CRediT authorship contribution statement

Muhammad Waqas: Writing – original draft, Methodology, Investigation, Data curation, Conceptualization. **Usa Wannasingha Humphries:** Writing – review & editing, Supervision, Project administration, Investigation, Conceptualization. **Angkool Wangwongchai:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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List of Abbreviations

NWP	Numerical Weather Prediction
AGCMs	Atmospheric General Circulation Models
OGCMs	Ocean General Circulation Models
AOGCMs	Atmosphere-Ocean General Circulation Models
GCMs	Global Climate Models
ESMs	Earth System Models
CMIP6	Coupled Model Intercomparison Project Phase 6
NEMO	Nucleus for European Modeling of the Ocean
HYCOM	Hybrid Coordinate Ocean Model
ENSO	El Niño–Southern Oscillation
AMO	Atlantic Multidecadal Oscillation
PDO	Pacific Decadal Oscillation
CESM	Community Earth System Model
CFSv2	Climate Forecast System version 2
HPC	High-Performance Computing
NMME	North American Multimodel Ensemble
AI	Artificial Intelligence
ML	Machine Learning
PIML	Physics-Informed Machine Learning
SST	Sea Surface Temperature
CNN	Convolutional Neural Network
MJO	Madden–Julian Oscillation
CAM	Community Atmosphere Model
POP	Parallel Ocean Program
CLM	Community Land Model
CICE	Community Ice Code
CPL	Coupler (e.g., CPL7 used in CESM)
MOM	Modular Ocean Model
LSM	Land Surface Model
WRF	Weather Research and Forecasting model
ARW	Advanced Research WRF
NMM	Nonhydrostatic Mesoscale Model
IFS	Integrated Forecasting System
GFS	Global Forecast System
FV3	Finite-Volume Cubed-Sphere Dynamical Core
PINN	Physics-Informed Neural Network
GAN	Generative Adversarial Network
DMDU	Decision Making under Deep Uncertainty
RASM	Regional Arctic System Model
MPAS	Model for Prediction Across Scales
RCM	Regional Climate Model
LAM	Limited Area Model
ERA5	ECMWF Reanalysis 5
MERRA-2	Modern-Era Retrospective analysis for Research and Applications, Version 2
GFDL	Geophysical Fluid Dynamics Laboratory
GFDL-CM4	GFDL Climate Model Version 4
EC-Earth	European Consortium Earth-system model

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MPI-ESM	Max Planck Institute Earth System Model
UKESM	UK Earth System Model
CNRM-ESM	Centre National de Recherches Météorologiques Earth System Model
FIO-ESM	First Institute of Oceanography Earth System Model
AWI-ESM	Alfred Wegener Institute Earth System Model
CWFR	Canonical Workflow Framework
GEFSv12	Global Ensemble Forecast System Version 12
SHAP	SHapley Additive exPlanations
LIME	Local Interpretable Model-Agnostic Explanations
GradCAM	Gradient-weighted Class Activation Mapping
AIFS	AI Forecasting System (by ECMWF)

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